Online Resource 1: List of samples taken in the field and investigated in this study.

Sample	Rock Type	Description
16B06	**Massive sulphide**	60-70 vol% pyrite, 5-15 vol% sphalerite, 2-5 vol% galena, 2 vol% tetrahedrite, 5-10 vol% barite, 2-5 vol% quartz. Pyrite is highly fractured showing numerous inclusions. Textures are framboidal (2 vol%, 20-40 µm), subhedral (200-600 µm) and euhedral (as inclusions in sphalerite; <5 µm). Mineral inclusions (<5-10 µm) are galena, sphalerite, rarer tetrahedrite. Sphalerite is mostly subhedral. Grain sizes range from 20 - 400 µm. Galena is subhedral to anhedral. Grain sizes are 10-50 µm. Tetrahedrite is subhedral. Grain sizes are 50-150 µm Barite is anhedral and irregularly distributed throughout the sample. Grain sizes are variable with the majority being 100-600 µm. Quartz is anhedral with grain sizes of 50-200 µm. The sample is cut by two prominent veins. One consisting of barite, the other of a quartz-barite mixture.
16B07	**Massive sulphide**	65-75 vol% pyrite, 5-15 vol% sphalerite, 2-5 vol% galena, 1-2 vol% tetrahedrite 10-15 vol% barite, 2-5 vol% quartz. Pyrite appears fractured and shows numerous inclusions. Pyrite textures are framboidal (2 vol%, 20-40µm), subhedral-euhedral (40-600 µm) and euhedral (<5µm). Mineral inclusions (<5-10 µm) are galena, sphalerite, rarer tetrahedrite. Galena is subhedral to anhedral. Grain sizes are 20-100 µm. Sphalerite is mostly subhedral. Grain sizes range from 20 - 500 µm. The sample is cut by a large barite vein which is barren of any sulphides.
16B09_2	**Massive sulphide**	75-85 vol% pyrite, 10-15 vol% sphalerite, 2-5 vol% galena, 5 vol% tetrahedrite, 2-5 vol% barite, 2-5 vol% quartz. Finer grained than other massive sulphide samples. Pyrite is fractured and shows numerous inclusions. Pyrite textures are framboidal (2 vol%, 20-40 µm), subhedral-euhedral, (40-600µm) and euhedral (<5 µm, closely associated with other sulphides). Mineral inclusions (<5-10 µm) are galena, sphalerite, rarer tetrahedrite. Galena is subhedral to anhedral. Grain sizes are 20 - 150 µm. Sphalerite is mostly subhedral. Grain sizes range from 20 - 500 µm.

16B10	**Volcaniclastic sandstone**	Chloritic-sericitic, clast-supported, crystal-rich volcanic sandstone. The matrix consists of sericitized plagioclase. Clast grains are dominated by quartz (100-300µm; 5-10 vol%) and oxide minerals (200-600 µm; 5-10 vol%). Other clasts are extensively chloritized and cannot be clearly identified (possibly pyroxene and/or amphibole).
16B11	**Volcaniclastic mudstone**	Chloritic-sericitic, matrix-supported, volcanic mudstone. The matrix consists mainly of plagioclase (50-60 vol%) and subordinate orthoclase, both sericitized. Crystal shapes further indicate clasts of chloritized pyroxene (30-40 vol%) and minor amounts of amphibole (<5 vol%). Small amounts of titanite and quartz can be found (<2 vol%).
16B12	**Barite**	Barite from barite vein within the massive sulphides
16B13	**Coherent alkali basalt**	Porphyritic texture, with fine-grained matrix and microphenocrysts; all minerals are extensively altered; cleavages and crystal shapes are somehow oblique. Feldspar is recognized by its lathlike crystals and is strongly sericitized. Other phenocrysts cannot be clearly identified. Where preserved, crystal shapes suggest amphibole and pyroxene in about equal proportions. Microprobe data delivers unclear signals due to irregular substitution with chlorite and other hydroxide-bearing minerals. Minor amounts of titanite and quartz can be found (<2 vol%) and Fe-chromites (<2 vol%). Given the strong alteration, an estimation of mineral distribution is not possible.
16B14	**Hydrothermally altered sand/siltstone**	95 vol% quartz, 1-2 vol% muscovite, single crystals of zircon (<1 vol%) and rutile (<1 vol%). Hydrothermal alteration led to mineralization of melanterite ($FeSO_4 \times 7H_2O$) and barite veining (2-5 vol%). The amount of melanterite is impossible to estimate, but it is finely dispersed in most of the sample. Located stratigraphically above the ore mineralization; more distant than sample 16B15

16B15	**Hydrothermally altered sediment**	Occurs in the field as sand/siltstone and adjacent to the main massive sulphides; completely overprinted by pyrite and barite. 60-70 vol% pyrite, 20-30 vol% barite, 5-10 vol% quartz. Occasional sphalerite and galena («1 vol%). Pyrite is anhedral and strongly fractured. Grain sizes are 50-600 µm. Also present are framboidal pyrites (1-2 vol%, <20 µm). Barite is intergrown with quartz and is deposited in an irregular network of fractures. - Located stratigraphically above the ore mineralization; below 16B14.